

SUMIT YADAV

AI Researcher | Interpretability & Safety | Applied NLP & Agentic Systems

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SUMMARY

AI researcher and engineer working at the intersection of **mechanistic interpretability**, **AI alignment/safety**, and **low-resource NLP**. My research studies how safety-aligned LLMs fail silently, through over-refusals, geometric misrepresentation, and surface-level triggers, and how the internal structure of representations can be understood and steered to make models safer and more reliable; this work has been accepted at **ACL 2026 (CORE A*)**. Alongside research, I design production-grade AI systems: **Zero-Trust** oversight for autonomous agents, **RAG** pipelines over vector databases, self-hosted LLM serving, and security tooling for the **Model Context Protocol**. I am an active Kaggle competitor and open-source contributor (creator of *maiBERT*, the first BERT for Maithili), and I care about extending interpretability and safety to multilingual and underserved communities.

TECHNICAL PROFICIENCY

Interpretability & Alignment: representation & activation steering, mechanistic interpretability, linear probes / CKA, representation geometry, effective dimension, over-refusal mitigation, adversarial probing, insight.

LLMs & NLP: LLMs (Llama 3, Claude, GPT-4o), fine-tuning (PEFT/LoRA), RLHF, RAG pipelines, multi-agent systems, multilingual / low-resource NLP, tokenizer training, embedding alignment.

Computer Vision & Audio: YOLOv8, DINOv2/v3, CLIP, SigLIP, SAM/SAM2, Qwen-VL, Perch / spectrogram models, Mamba hybrids.

AI Security & Agentic Systems: Model Context Protocol (MCP), MITRE ATT&CK mapping, prompt-injection & tool-poisoning defense, Zero-Trust architecture, Guardrails, interactive permission systems.

Architecture: Hexagonal (Ports & Adapters), Domain-Driven Design, microservices, event-driven (SSE), REST/GraphQL APIs.

Vector DBs: Qdrant, ChromaDB, Pinecone, Weaviate (hybrid search, HNSW).

Languages & Tools: Python (expert), C++, C, SQL, Bash, JavaScript; PyTorch, TensorFlow, vLLM, LangChain, LlamaIndex, FastAPI, Docker, Kubernetes, GitHub Actions.

PROFESSIONAL EXPERIENCE

Astha.ai

AI Researcher, AI Security & Agentic Systems

USA (Remote)

May 2025 – Present

- **SAFE-MCP scanner (primary author):** built the MCP vulnerability scanner for agentic AI systems — tool poisoning, injected tool descriptions and related SAFE-MCP techniques — pairing pattern-based detection with LLM analysis across **78+** MITRE ATT&CK-mapped techniques; added a statistical analysis mode that makes non-deterministic LLM verdicts reproducible.
- **LLM-powered code scanner:** hardened the LLM integration layer against prompt injection (canary tokens, untrusted-input escaping, response validation) and built **tree-sitter** semantic code chunking for stronger context, plus observability, caching and quota enforcement; raised test coverage **40% → 75%** and cleared the strict type-check backlog **to zero**.
- **safe-k8s, Kubernetes compliance agent (lead engineer):** Go service running inside customer clusters and in the SaaS platform, with authenticated comms, metrics, heartbeats and evidence trails; built cluster-aware scan planning where an LLM refines selection from a catalog of **~470** compliance-control skills, and repeated-run consensus with fixed sampling that makes verdicts deterministic.
- **Cloud compliance agents:** shipped scanners for **AWS, GCP and Google Workspace (251 checks)** as Cloud Run Jobs, adding organization-boundary enforcement, readiness gates and coverage declaration so a partial scan is never reported as complete.
- **Platform & GitOps:** own the dev → staging → production promotion path for a multi-tenant AI security & compliance SaaS on **GKE/Istio** using Pulumi, Helm and ArgoCD with cosign-signed, digest-pinned images; primary reviewer on the core infra repos, enforcing SOC 2 change control with **zero admin-merge bypasses**; root-caused a two-day outage from a stale API-key rotation and shipped its P0 detector as IaC. **800+** commits across **30+** repos (Python, Go, Rust, TypeScript).

AMNIL Technologies

AI Engineer, RAG & Infrastructure

Lalitpur, Nepal (Hybrid)

Jun 2024 – May 2025

- **Advanced RAG:** built a hybrid-search (sparse + dense) RAG system on **Qdrant** with recursive query decomposition, improving retrieval accuracy by **35%** on multi-hop, domain-specific questions.
- **Self-hosted LLM serving:** deployed open models (Llama 3, Mistral, Qwen) on **vLLM**, tuning PagedAttention / KV-cache to cut latency by **40%** vs. standard Hugging Face pipelines.
- **Guardrails & evaluation:** integrated guardrails to enforce topical boundaries and built an automated **LLM-as-a-Judge** framework to benchmark response quality.

GradeUp Educations

Kathmandu, Nepal

Data Team Lead

Jan 2022 – Jun 2024

- **Low-resource document AI:** led OCR and document parsing for Nepali and Maithili, overcoming data scarcity via synthetic Devanagari training data; built an automated grade-evaluation system.
- **Semantic search:** fine-tuned embedding models with contrastive learning for educational content, improving relevance for non-English queries; shipped learning agents / chatbots.
- **Leadership:** defined the data roadmap and led a small team across the ML lifecycle from ideation to deployment.

DeepLearning.AI

Global (Remote)

GAN Specialization Mentor

Aug 2021 – Present

- Technical mentor for the Generative Adversarial Networks Specialization, loss functions (Minimax, Wasserstein), training stability, and architectures (DCGAN, CycleGAN), supporting students worldwide.

RESEARCH & PUBLICATIONS

Peer-reviewed and preprint work at the intersection of AI safety, interpretability, and low-resource NLP. Full list on [Google Scholar](#).

SafeConstellations: Mitigating Over-Refusals in LLMs Through

ACL 2026 Main (CORE A*)

Task-Aware Representation Steering | AI Safety, Representation Steering

- LLMs trace consistent, task-specific “constellation” trajectories in embedding space that shift predictably between refusal and non-refusal states.
- Proposed an inference-time, task-aware trajectory-shifting method that reduces over-refusals by up to **73%** with minimal utility loss and no retraining, accepted at **ACL 2026 Main Conference**. (with U. Maskey, M. Dras, U. Naseem.) [\[ACL Anthology\]](#) [\[arXiv\]](#)

Your Intervention Was Too Narrow: Mechanism Width as a Precondition for Interpretability-Driven Discovery (first author) |

NeurIPS 2026 Workshop

Interpretability, Causal Intervention

- The field’s standard intervention unit (one direction, one layer, one attention head) recovers only about **10%** of the achievable causal effect across three unrelated testbeds: rank-1 steering in DINOv2 (0.111), single-layer ablation in Llama-3.1-8B (0.096) and single-head sufficiency in GPT-2’s IOI circuit (0.086). Mechanism width k^* is **measurable before intervening**, and matching it recovers **95–100%**; a null at standard width is therefore an **underpowered test, not evidence of absence**. (with B. Joshi.) [\[NeurIPS\]](#) [\[project\]](#) [\[preprint\]](#)

Geometric Phases of Mechanism Formation in Neural Networks

GLOW @ IJCAI-ECAI 2026

(sole-authored) | Interpretability, Training Dynamics

- Using linear probes and CKA across dense training checkpoints, shows classification mechanisms form **output-layer-first** within the first ~5% of training (Cohen’s $d = 3.68$); the deep-first pattern holds in from-scratch LLM pretraining (GPT-2 Small, SmolLM2-135M) and reproduces on public Pythia and OLMo-2 checkpoints.

On the Relationship Between Representation Geometry and

Pre-Print 2026

Generalization (sole-authored) | Representation Learning, Information Geometry

- Establishes that a label-free geometric metric, **effective dimension**: strongly predicts generalization across vision and language (partial $r = 0.75$ over 52 ImageNet models, 13 architectures; carries to NLP encoders and decoder-only LLMs).
- Shows the link is causal: degrading the geometry hurts accuracy while PCA cleanup preserves it, a diagnostic relevant to distribution-shift detection and robustness. [\[arXiv\]](#)

MaiBERT: A Pre-training Corpus and Language Model for

LoResLM @ EACL 2026

Low-Resourced Maithili | NLP, Low-Resource Languages

- First monolingual BERT for Maithili (~50M speakers), pre-trained on a custom-curated corpus with a Masked Language Modeling objective.
- Achieves **87.02%** news-classification accuracy, outperforming multilingual baselines (MuRIL, NepBERTa), addressing fairness and access for underrepresented communities. [\[ACL Anthology\]](#) [\[Hugging Face\]](#)

How Far Can a Single Vector Carry a Language? (*first author*) | Under review, Machine Learning (Springer) *Multilingual Steering*

- With **no fine-tuning**, steers a model's Hindi representations toward Devanagari languages it does not support (Nepali, Maithili, Bhojपुरi). Finds a shared **hourglass geometry** across depth and an **adherence–fluency frontier** that bounds single-direction steering: adherence rises then collapses while fluency declines. Separability, not proximity, predicts which targets are steerable; source choice tracks linguistic proximity rather than resource level (Hindi 3.23 vs English 1.37).

Revolutionizing Currency Security with YOLOv8 | *Computer Vision, Object Detection*, J. Bus. Econ. Stud. 2024

- Applied YOLOv8 to counterfeit Nepali banknote detection on a custom dataset, achieving **0.986** true-positive recall on the reverse face, enabling a practical mobile verification solution. [\[DOI\]](#)

Machine Learning Analysis of Tirhuta Lipi | *Script Recognition, Cultural Heritage* 2023

- Character-level analysis and recognition for the historic Tirhuta (Mithilakshar) script, building the data and modeling groundwork for digitizing an endangered writing system.

Support Vectors for Imbalanced Text Classification | *Machine Learning Theory* 2023

- Showed that for highly imbalanced datasets with 100+ classes, optimized SVMs with TF-IDF often outperform deep models on training efficiency and F1-score stability.

COMPETITIVE ML & OPEN SOURCE

Kaggle & Data-Science Competitions (2025–2026) | *20+ models across 5 challenges* Global

- **BirdCLEF+ 2026**: acoustic species identification from Pantanal wildlife audio. Trained DINOv2-based spectrogram classifiers (224px), convolutional baselines, and Google Perch 2.0 transfer learning; earned a **Bronze medal** with a kNN-probe on Perch embeddings.
- **CSIRO Image2Biomass**: pasture above-ground biomass estimation from images + metadata. Built hybrid vision models: DINOv3 + Mamba (3-head, improved, SigLIP-Mamba variants), DINOv3 + SAM, and DINOv2 baselines.
- **Vesuvius Challenge (Ink Detection)**: 2.5D/3D ink segmentation on carbonized Herculaneum scrolls. Explored DINOv2 2.5D, 3D UNet variants with topological features (multi-GPU), and TransUNet + DINOv2 ensembles.
- **Scientific Image Forgery Detection (Recod.ai/LUC)**: copy-move detection & segmentation in biomedical images. Broad architecture sweep: CNN–DINOv2 hybrids, ConvNeXt + progressive decoders, EfficientNet-B0, YOLO + SAHI, fine-tuned SAM2 / Qwen3-VL, and steered vision models across multiple ensembles.
- **Community**: founder of **NPL Coders**, organizing national-level data-science hackathons to grow Nepal's AI ecosystem.

Open Source | *GitHub: rockerritesh* Open Source

- **maiBERT-TF**: first open BERT pre-trained for Maithili, used by the linguistic-research community. **Maithili speech**: open Whisper-tiny ASR and MMS-TTS TTS with live HF Spaces demos.
- **SAFE-MCP**: maintainer of the community security framework for the Model Context Protocol; authored ATT&CK-style detection rules and PoCs (server enumeration, tool-chaining pivots, multimodal prompt injection).
- **Telebot-Claude-Bridge**: permission-gated, human-in-the-loop remote control of the Claude Code CLI via Telegram.
- **Removing the “Workspace” Layers of Llama**: interpretability study (write-up + code): a decodable-yet-entangled middle band (L6–L16) in Llama-3.1-8B whose removal collapses reasoning while shallow recall survives; probes + reversible layer-ablation, scaling to 405B via nnsight/NDIF. [\[code\]](#)

- **World Models (toy experiments):** Dreamer-style RSSM world models built from scratch for continuous and visual control; agents trained purely in imagination transfer to the real task (CartPole 500/500). [\[code\]](#)

KEY ENGINEERING PROJECTS

Telebot-Claude-Bridge | *Agentic tooling, Python + Telegram API* 2025–2026

- Production bridge for remote control of the **Claude Code CLI** entirely from Telegram: bot ↔ HTTP webhook ↔ tmux-hosted session with hooks relaying tool calls, stops, and permission prompts in real time.
- Interactive Allow / Deny / Allow-All permission buttons per tool call (zero-trust), multi-session switching with auto-recovery, and code-block-aware message splitting. [\[GitHub\]](#)

Nepali Chat-with-Doc & Localization Suite | *RAG, Unicode, Font Conversion* Production

- Full-stack RAG chatbot optimized for Nepali, with a real-time Preeti↔Unicode conversion layer that lets the LLM process legacy government documents; reflexive agents self-correct on poor retrieval.

Agents.ai & Semantic Router | *Vector DB, Multi-Agent Systems* 2024

- Intent-aware router that selects the best expert agent (research / coding / data) by semantically matching queries against capability profiles stored as embeddings in ChromaDB.

IRB Robotic Arm | *IoT, Signal Processing, C++* 2020

- Image-Recognition-Based robotic arm for touchless medical assistance (UN SDG3): TensorFlow gesture recognition with Arduino actuator control.

EDUCATION

Pulchowk Engineering College, IOE, Tribhuvan University Kathmandu, Nepal
Bachelor of Computer Engineering 2019 – 2024

- B.E. thesis: “Evaluating Auto-Encoding Transformer Language Models for Maithili Text Classification”: my first Maithili masked language model and the foundation for maiBERT.
- Key subjects: Artificial Intelligence, Data Structures & Algorithms, Distributed Systems, Big Data Technologies, Network Security, Compiler Design.

SERVICE, MENTORSHIP & COMMUNITY

Reviewer: Mechanistic Interpretability Workshop, ICML 2026 (recognized as an **Outstanding Reviewer**).

Mentorship: technical mentor, DeepLearning.AI GANs Specialization (2021–present), supporting students worldwide.

Community building: founder of **NPL Coders**, organizing national data-science hackathons to grow Nepal’s AI ecosystem; maintainer and contributor across open-source NLP and AI-security projects.

Open science: public releases of maiBERT, Maithili ASR/TTS, and interpretability experiments with reproducible code and live demos.

HONORS & AWARDS

Best Paper Candidate, GLOW @ IJCAI-ECAI 2026 (sole-authored) · **Outstanding Reviewer**, Mechanistic Interpretability Workshop (ICML 2026) · **Winner**, GritFeat AI Hackathon (2023) · **1st Runner-Up**, Locus Dataverse (2023) · **1st Runner-Up**, Docsumo DataRush (2022)

Best AI Project, DELTA 3.0 (2023) · **Winner**, IT-Meet Image Challenge (2022) · **Winner**, LogPoint Capture The Flag (2022)

SELECTED CERTIFICATIONS

Deep Learning Specialization: DeepLearning.AI (Coursera).

Generative Adversarial Networks (GANs) Specialization: DeepLearning.AI (Coursera).

LANGUAGES

Maithili (native) · **Nepali** (fluent) · **Hindi** (fluent) · **English** (professional working proficiency).